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(12) **United States Patent**
Cuddeback et al.(10) **Patent No.:** **US 12,403,158 B1**(45) **Date of Patent:** ***Sep. 2, 2025**(54) **PHYTOCHEMICAL/ NUTRACEUTICAL
COMPOSITION FOR MULTIMODAL
PROPHYLAXIS AGAINST AND TREATMENT
OF VIRAL AND BACTERIAL INFECTION
AND INFLAMMATION**(71) Applicant: **David A Cuddeback**, Kirkwood, NY
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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.This patent is subject to a terminal dis-
claimer.(21) Appl. No.: **18/410,512**(22) Filed: **Jan. 11, 2024****Related U.S. Application Data**(60) Division of application No. 18/094,541, filed on Jan.
9, 2023, now Pat. No. 11,896,611, which is a division
of application No. 17/682,300, filed on Feb. 28, 2022,
now Pat. No. 12,186,341, which is a
continuation-in-part of application No. 16/848,393,
filed on Apr. 14, 2020, now abandoned.(51) **Int. Cl.***A61K 36/00* (2006.01)
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A61K 36/82 (2006.01)(52) **U.S. Cl.**CPC *A61K 33/30* (2013.01); *A61K 9/0053*
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A61K 36/537 (2013.01); *A61K 36/74*
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CPC *A61K 31/375*; *A61K 31/355*; *A61K 33/30*
See application file for complete search history.(56) **References Cited**

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Primary Examiner — Qiuwen Mi(57) **ABSTRACT**A phytochemical and nutraceutical composition and method
for protection against a wide spectrum of viral and bacterial
infections, including Covid-19, and for treatment of estab-
lished infection and infectious inflammation. The composi-
tion includes a novel combination of vitamin, mineral,
nutraceutical and phytochemical supplements. The composi-
tion of supplements and method may be compounded as a
pill, tablet, powder, capsule or liquid be taken orally one or
more times per day, and parenteral including nasal sprays
and inhalers. Vitamin C and vitamin E along with zinc
complexed with pyridione or citrate are used in conjunction
with nutraceuticals and phytochemicals provided, respec-
tively, as immune boosters and antiviral agents along with
anti-inflammatory nutraceuticals and phytochemicals.**11 Claims, No Drawings**

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1

**PHYTOCHEMICAL/ NUTRACEUTICAL
COMPOSITION FOR MULTIMODAL
PROPHYLAXIS AGAINST AND TREATMENT
OF VIRAL AND BACTERIAL INFECTION
AND INFLAMMATION**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This Application is a continuation utility patent application from divisional of Ser. No. 18/094,541 filed Jan. 9, 2023, which is a divisional application Ser. No. 17/682,300 filed Feb. 28, 2022, which is a continuation-in-part of utility application Ser. No. 16/848,393 filed Apr. 14, 2020 by the same inventors.

BACKGROUND OF THE INVENTION

Field of the Invention

This invention relates to phytochemical/nutraceutical compositions, products and methods that include multimodal prophylaxis against and treatment of viral and bacterial infections and inflammations, especially in human beings.

Description of Related Art

Nutraceuticals are: A) Dietary Supplements, such as vitamins, minerals, herbs, amino acids, enzymes, organ tissues or metabolites and, B) Functional Foods, defined in a position statement by the Academy of Nutrition and Dietetics: Functional Foods. Kristi Crowe; Coni Francis, J Acad Nutr Diet. 2013; 113:1096-1103, as “whole foods along with fortified, enriched, or enhanced foods that have a potentially beneficial effect on health when consumed as part of a varied diet on a regular basis at effective levels based on significant standards of evidence.” Examples would be “the antioxidant vitamins in orange juice, isoflavones in soy-based foods, and prebiotics and probiotics in yogurt.” In all such instances, the antioxidant or other functional ingredient is contained within an intact, whole food.

Phytonutrients, defined by the NIH/National Cancer Institute, “are organic components of plants thought to promote human health. Fruits, vegetables, grains, legumes, nuts and teas are rich sources of phytonutrients. Unlike the traditional nutrients (protein, fat, carbohydrates, vitamins, and minerals), phytonutrients are not ‘essential’ for life, so some people prefer the term Phytochemical.”

Phytochemicals, within the parent plant, help the plant resist microbial attack by fungi, bacteria and viruses. Phytochemicals are further defined by Liviuta Budisan et al; Int J Mol Scvi. 2017 June; 18(6): 1178 as; “natural compounds synthesized as secondary metabolites in plants, representing an important source of molecules with a wide range of therapeutic applications.” Budisan et al. explains further that: “these natural agents are important regulators of key pathological processes/conditions, including cancer, as these natural agents are able to modulate the expression of coding and non-coding transcripts with an oncogenic or tumor suppressor role.”

Phytochemicals are also known to have multiple antiviral effects in vitro and multiple anti-inflammatory effects in vitro and in vivo.

For phytochemicals to be most effective in their use, they must be extracted in concentrated form from the parent plant.

2

Chemical classifications of the family of phytochemicals are all slightly different.

Polyphenols are phytochemicals, found largely in fruits, vegetables, tea, coffee, chocolates, legumes, cereals, and beverages. There are over 8000 polyphenols identified in nature and their main function is antioxidant and other significant health benefits.

Many effective drugs have come originally from phytochemical sources. A selected few examples follow: Willow bark was used more than 3500 years ago as an analgesic and antipyretic, although those terms were not used at the time. About 1897, Felix Hoffman discovered its active component to be the phytochemical salicin, which inspired his synthesis of salicylic acid (aspirin). Atropine first came from the *belladonna* plant (aka nightshade) and is now used to decrease respiratory secretions during surgery, to treat poisoning by certain cholinergic nerve agents and as the first line treatment for symptomatic bradycardia in the emergency department. Paclitaxel was isolated from the English Yew tree in 1971 and has since been used to treat ovarian, breast, and non-small cell lung cancer. In 1928, Alexander Fleming discovered the first antibiotic, penicillin, in mold of the genus *Penicillium*.

There is renewed interest in phytochemicals because of more recent research that has demonstrated antiviral and antibacterial efficacy of various phytochemicals in numerous in vitro experiments and especially now because of the prophylactic and therapeutic quandary presented by the SARS COV-2 pandemic. The COVID-19 oral antiviral pills recently introduced by Pfizer, Paxlovid (ritonavir), and by Merck, Lagevrio (molnupiravir), were first discovered in in vitro research, then confirmed in in vivo (human) clinical trials. It is a common research pattern that positive results in vitro often become positive results in vivo (humans and animals).

Phytochemicals are not food, vitamins or minerals per se. They are plant chemicals with multiple, evolved non-nourishment purposes, particularly plant defense. Of note, their molecular structures can bear resemblance to laboratory chemicals invented for human defense.

Before modern pharmaceuticals, mankind depended on natural substances for medical purposes which included, but were not limited to, relieving pain, enhancing sexual and other performance, enhancing sleep, enhancing bowel function, healing wounds, decreasing respiratory difficulty, stopping blood loss, halting the progress of illness and augmenting general wellbeing. Such natural substances included, but were not limited to, vegetables, herbs, roots, flowers, tree barks and certain animal organs and tissues. A key target of the use of natural substances was abatement of the combination of pain, swelling and redness in any area of the body, a combination which by the 16th century came to be known as inflammation. At that time and before, inflammation was an important target for treatment with natural substances because it was well recognized that inflammation could progress to worse, morbidity or death. Although allergic or purely traumatic injury can also cause inflammation, in the 19th century it was discovered that bacterial infection was a chief and more dangerous cause. In the 20th century, knowledge of the causes of inflammation expanded to include viral and fungal infections.

In our time, natural substances have evolved to include isolated atoms known as minerals, isolated biomolecules known as vitamins and purified extracts from any of the above mentioned, historical sources of natural substances. There is a large variety of such modern natural substances, which in modern pill, tablet, liquid, powder or capsule form

are now known as dietary or health supplements, or nutraceuticals. Nutraceuticals and phytochemicals are currently used for essentially the same array of purposes as those throughout history, but now also for fortification of the immune system against primary acquisition of infection, i.e. prophylaxis, and for fortification of the immune system's ability to defeat infection once it is established.

Modern antibacterial and antiviral vaccines are generally effective against only the bacterial or viral pathogen from which they were pharmaceutically derived, not a range of bacteria or viruses. Modern (antibacterial) antibiotics for use after infection has already occurred are typically quite effective against some range of bacteria. Historically however, antiviral medications for use after viral infection has already occurred, work against only a specific virus, have minimal and even questionable effectiveness and also have severe time constraints on their uses.

A small variety of available supplements are targeted at overall fortification of the innate immune system, with the intent to prevent or mollify a variety of viral and bacterial infections such that no infection occurs, or that its symptoms and physiologic effects may be restricted to the minimal and the brief. These supplements go by such labels as Immune Boost, Immune Health and others, and many, if not most, are composed of a single active ingredient, and none claims both immune fortification and inclusion of a known anti-inflammatory substance.

What is needed is a novel and optimal combination of immune fortifying ingredients, each of which works by a different mechanism to prevent, mollify and treat a wide range of viral and bacterial infections (hereafter termed wide-spectrum anti-infectious), and another ingredient or ingredients which is/are anti-inflammatory, in the event that some degree of infectious inflammation becomes established.

Since whole-food nutraceuticals are an inefficient and minimally effective source of immune-fortifying, antiviral and anti-inflammatory ingredients, what is needed is a novel and unique combination of extracted phytochemicals whose wide-spectrum prophylactic effect against infection is compounded by its ability to treat already established infection and infectious inflammation, such that its overall prophylactic and therapeutic effects synergistically equal more than the sum of its parts, as does the fortuitous combination of vitamin C+thiamine+hydrocortisone in reversal of sepsis, a severely life-threatening condition which can follow routine infection with a virus or bacterium.

Phytochemical extracts do not exist in nature. Further, any extract of a plant contains not one phytochemical but an array of phytochemicals. The above combination of vitamin C+thiamine+hydrocortisone used for sepsis is referred to as a clinical cocktail. Clinical cocktails are little used in medicine. Instead, the tradition in Western medicine is one clinical drug for one clinical problem, which is not always successful. The purpose of a clinical cocktail is to overwhelm the clinical problem by causing multiple corrective effects at once. To protect against viral infection, we propose a combination three nutraceuticals that support the innate immune system (zinc, vitamin C and vitamin E) plus an unnatural combination (cocktail) of seven antiviral phytochemical extracts from seven different plants, wherein each extract itself contains its own sub-array of antiviral phytochemicals. Representative literature references citing the in vitro antiviral and/or antibacterial efficacy of the phytochemical extracts we propose include those for extracted fulvic acid, *Radix bupleuri* extract, *Uncaria tomentosa*

extract, pomegranate extract, black elderberry extract, sage extract, and *Echinacea* extract.

The invention utilizes a phytochemical cocktail approach to prevent or treat health-threatening inflammation caused by an established viral or bacterial infection. To protect against infectious inflammation, the invention includes a method and a composition having a combination of two anti-inflammatory nutraceuticals (omega-3 fatty and -lipoic acid) plus an unnatural combination (cocktail) of eight anti-inflammatory phytochemical extracts from eight different plants, wherein each extract itself contains its own sub-array of anti-inflammatory phytochemicals. Representative literature references provided in an information disclosure statement citing the in vitro and/or in vivo anti-inflammatory efficacy of the phytochemical extracts contained in the invention include those for *Boswellia serrata* extract, maritime pine bark extract, *Uncaria tomentosa* extract, *Commiphora myrrha* extract, resveratrol, alpha-lipoic acid, curcumin extract with piperine, green tea extract, Omega-3 fatty acids, and cannabidiol.

Among the obvious benefits of virology research is the elucidation of viral structures, and the mechanisms of virus-to-host-cell-attachment, virus entry and replication and viral budding. Among the obvious benefits of other medical research is elucidation of the numerous mechanisms of cell function and the numerous mechanisms of normal versus maladaptive inflammation. The nutraceuticals contained in the proposed antiviral cocktail in the invention method and composition support numerous facets of the innate immune system, and the phytochemicals contained therein are known to combat many or most of the mechanisms of viral infection and replication.

The nutraceuticals and phytochemicals contained in the anti-inflammatory cocktail described herein are known in the literature to inhibit or block a key set of chemical mediators of inflammation (e.g. cytokines,) that cause serious viral inflammation, and notably those that cause cytokine storm in COVID-19. These mediators include IL-1B, IL-6, NF-kappa B, leukotrienes, TNF-alpha and the enzyme, human leukocyte elastase, believed to cause lung tissue destruction in emphysema.

In addition to cytokines, oxygen free radicals are also known to be key players in inflammatory processes. So too, inflammatory processes in the vascular endothelium cause endotheliitis in COVID-19, which enables thrombogenesis and which in turn creates multiple clots that cause stroke, heart attack and renal failure. Another facet of the invention anti-inflammatory phytochemical cocktail, with its inherent subarrays of other phytochemicals, is that many of the phytochemicals contained therein also have antioxidant and/or antithrombotic effects.

Still another facet of the invention method and compositions enclosed, is that desired effects, whether antiviral, anti-inflammatory, antioxidant or antithrombotic, should be potentiated by the effects of each component compounding the effects of the others. This has been found for pomegranate rind extract and punicalagin when co-administered with zinc in its virucidal activity against both herpes simplex virus (HSV) and acyclovir-resistant HSV. Of the multiple, possible mutual potentiations among our phytochemicals, synergy between two or more is also possible, synergy in which the combined effect of two or more is significantly greater than the sum of their individual effects. Synergy has been found for the anti-inflammatory effect of the co-administration of *Commiphora myrrha* extract with *Boswellia* extract.

A common feature of laboratory-invented antiviral medications (e.g. oseltamivir & zanamivir) used against influenza virus is that strains of influenza (e.g. H1N1 and H3N2) rapidly develop resistance to the medication and thereafter become ineffective in certain populations. A curiosity of several of our proposed anti-infective phytochemicals is that even after multiple passes in vitro, the viruses against which they were being tested failed to develop resistance to the phytochemicals, whereas the same number of in vitro passes caused significant viral resistance against laboratory-invented antiviral medications. There is as yet no explanation for this advantage of phytochemicals.

By and large, research has not tested the combination of even two co-administered phytochemicals for their anti-infectious or anti-inflammatory effects, let alone a cocktail of 3 or more. An unexpected result thereof, or perhaps a foregone conclusion, should be that since each of these phytochemical extracts has been shown to be effectively antiviral against one or more viruses in vitro, then a cocktail of all seven extracts should be even more antiviral against any one virus, and effectively antiviral against a much wider range of different viruses.

SUMMARY OF THE INVENTION

A novel combination of vitamin, mineral and herbal or other natural, organic supplements (phytochemicals) for protection against a wide spectrum of viral and bacterial infections, and for treatment of established infection and infectious inflammation. The composition of vitamin, mineral and phytochemical supplements is to be compounded as a pill, tablet, powder, capsule or liquid to be taken orally one or more times per day, or as parenteral liquid, that includes nasal sprays and inhalers, as possible routes of administration beyond the oral.

Among the possible innate immune system-supporting vitamins to be used are vitamins C and E. The mineral to be used is zinc complexed with citrate, pyridoxine, or the amino acid methionine, or other cofactor, in order to increase zinc's bioavailability.

Phytochemicals provided herein as immune boosters and/or antiviral agents to be combined as a composite in one vehicle include *Echinacea purpurea* extract, extracted fulvic acid, black elderberry extract, sage extract, Cat's Claw extract, pomegranate extract, *Radix bupleuri* extract, and cannabidiol (CBD).

The anti-inflammatory phytochemicals and nutraceuticals to be used are *Boswellia serrata* extract, maritime pine extract, Cat's Claw extract, green tea extract, resveratrol, curcumin with piperine, alpha lipoic acid, *Commiphora myrrha* extract, cannabidiol (CBD) and the omega-3 fatty acids, DHA and EPA. Methods of using these compositions of vitamins, minerals, herbs and other organic substances and compositions are also disclosed herein. The amount of each individual supplement substance may be varied significantly.

An object of the invention is to provide compositions of phytochemical extracts and nutraceuticals to fortify the human or companion animal immune system in a way that prevents a wide spectrum of known or new strains of viruses and bacteria from starting a new infection in the host.

Another object of the invention is to provide compositions of phytochemical extracts and nutraceuticals to fortify the human or companion animal immune system in a way that permits nascent viral or bacterial infection, once established, to be treated and defeated before the infection can cause

serious and/or widespread systemic symptoms and effects, or to minimize such symptoms to a life non-threatening level.

Another object of the invention is to provide compositions of phytochemical extracts and nutraceuticals that prevent or limit inflammation, especially in the lungs, in which virally or bacterially induced inflammation causes arteriolar and capillary leakage and resultant pulmonary edema, excess mucus secretion, possible severe bronchoconstriction and, in the worst case, acute respiratory distress syndrome (ARDS) with airway collapse, any of which phenomena can compromise pulmonary function to the point of fatal respiratory failure.

Another object of the invention is to provide compositions of anti-infective (antiviral and antibacterial) phytochemical extracts and nutraceuticals, already established to be safe for consumption that can be taken electively before anticipated viral exposure, prophylactically on a daily basis if desired, or therapeutically at the start of early viral symptoms.

Another object of the invention is to provide compositions of anti-inflammatory phytochemical extracts and nutraceuticals, already established to be safe for consumption that can be taken electively before anticipated viral exposure, prophylactically on a daily basis if desired, or therapeutically at the start of early viral symptoms.

Another object of the invention is to prevent or limit development of virally induced secondary bacterial infection/inflammation, or vice versa, especially in the lungs. An overall object of the invention is to help flatten the viral epidemic curve of yearly Influenza, and that of any new viral epidemic. Other features and advantages of the present invention will become apparent to those of skill in the art to consideration of the ensuing description and the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

For the purposes of this specification, the word "nutraceutical", refers to a food item, or a part of a food item, that offers medical health benefits, including prevention and/or treatment of disease. More particularly, a nutraceutical is a material that is:

- a dietary supplement containing a nutritive bioactive compound; or
- a biologically active process or process material derived from a vitamin, mineral, and herbal/organic supplements plant, fungus, and animal, or a portion thereof; where the precise composition of the biologically active process or unprocessed material may be undetermined. The word phytochemical refers to non-nutritive organic components of plants that are thought to promote human health. Unlike the traditional nutrients (protein, fat, carbohydrates, vitamins and minerals), phytochemicals are not essential for life, but have a wide range of therapeutic applications. Phytochemicals, within the parent plant, help it resist microbial attack by fungi, bacteria, and viruses.

Examples of a biologically active process material may include a finely chopped, powdered, pureed, or cooked material derived from plant or animal tissue, or an extract of plant or animal tissue. These agents are useful for accelerating improved immune function; and they include extracts from various plants and mixtures thereof.

A novel and unique combination of vitamin, mineral and herbal/organic supplements (phytochemicals and nutraceuticals) in one formulary forming one or more compositions

and methods for the purpose of fortifying the innate (cellular) and adaptive (hormonal) components of the immune system against bacterial and viral infection, and their propagation, transmission and effects.

Also included in the formulary compositions are two or more natural, herbal or other organic anti-inflammatory substances for the suppression of tissue inflammation, especially in the lungs. The formulary thereby provides multimodal nutraceutical and phytochemical protection against acquisition of infection and against deleterious effects of inflammation in the lungs. The formulary is available in both oral and parenteral forms, including nasal sprays and inhalers.

In a preferred embodiment, the formulary composition contains:

5 mg-50 mg zinc in any of its pyridione, citrate, gluconate, picolinate, acetate, sulfate or orotate forms, or with the cofactor methionine.

10 mg-2000 mg vitamin C

5 mg-1000 mg vitamin E

Two or more of the following wide-spectrum anti-infectious phytochemicals:

10 micrograms-2000 mg extracted fulvic acid

10 micrograms-1000 mg *Radix bupleuri* extract

10 micrograms-1000 mg *Uncaria tomentosa* extract

10 micrograms-1750 mg pomegranate extract

10 micrograms-2000 mg Black Elderberry extract or American Elderberry extract

10 micrograms-3200 mg Sage extract;

10 micrograms-1500 mg *Echinacea purpurea* extract

and two anti-inflammatory nutraceuticals:

10 micrograms-900 mg Omega-3 fatty acids, eicosapentanoic acid (EPA) and docosahexanoic acid (DHA).

10 micrograms-1200 mg Alpha-lipoic acid

and two or more of the following anti-inflammatory phytochemicals

10 micrograms-1800 mg *Boswellia serrata* extract, containing its most anti-inflammatory moiety, 3-acetyl-11-keto- β -boswellic acid (AKBA)

10 micrograms-750 mg Maritime pine bark extract

10 micrograms-400 mg *Uncaria tomentosa* extract

10 micrograms-800 mg *Commiphora myrrha* extract

10 micrograms-500 mg Resveratrol

10 micrograms-1200 mg Curcumin extract with piperine

10 micrograms-900 mg Green Tea extract

10 micrograms-680 mg Cannabidiol (CBD)

Any of the listed wide-spectrum anti-infectious nutraceuticals and phytochemicals permutated with any of the listed anti-inflammatory nutraceuticals and phytochemicals will confer five levels of anti-infectious protection, including white blood cell fortification and activation, virucidal-bactericidal activity, enhancement of antibody production, antioxidant activity and anti-inflammatory activity. The oral formulary, capsule or other, is enteric-coated to prevent breakdown by gastric acid and permit delivery of unaltered nutraceuticals and phytochemicals to the duodenum for absorption.

The method of the invention includes the steps of providing a patient with an antiviral and/or anti-inflammatory capsule or other oral form, parenteral administration, and nasal sprays and inhalers, containing a nutraceutical and phytochemical composition described herein to treat viral infections and inflammations resulting therefrom, once or more daily, in prescribed amounts, depending on the condition of the patient.

Although the present invention has been described in detail with particular reference in a preferred embodiment thereof, it should be understood that the invention is capable of other different embodiments, and its details are capable of modifications in various obvious respects. As is readily apparent to those skilled in the art, variations and modifications can be affected while remaining within the spirit and scope of the invention. Accordingly, the foregoing disclosure description is for illustrative purposes only and do not in any way limit the invention which is defined only by the claims.

What is claimed is:

1. A nutraceutical composition in capsule form for preventing or mollifying viral or bacterial infection, and also for preventing or mollifying viral or bacterial tissue inflammation, the composition comprising:

an effective amount of vitamin C;

an effective amount of vitamin E;

an effective amount of zinc;

effective amounts of wide-spectrum anti-infectious phytochemicals comprising extracted fulvic acid and at least one other anti-infectious phytochemical;

effective amounts of an anti-inflammatory nutraceutical comprising an omega-3 fatty acid, wherein the omega-3 fatty acid comprises eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), or both; and

effective amounts of anti-inflammatory phytochemicals comprising *Boswellia serrata* extract and at least one additional anti-inflammatory phytochemical.

2. The composition of claim 1, wherein the effective amounts of wide-spectrum anti-infectious phytochemicals comprise effective amounts of two or more of the following:

Radix bupleuri extract;

Echinacea purpurea extract;

sage extract;

pomegranate extract;

Uncaria tomentosa extract;

Black Elderberry extract; or

American Elderberry extract.

3. The composition of claim 1, wherein the effective amounts of anti-inflammatory phytochemicals comprise an effective amount of two or more of the following:

Pinus maritima bark extract;

Uncaria tomentosa extract;

Commiphora myrrha extract;

Resveratrol;

Curcumin extract with piperine;

green tea extract; and

Cannabidiol.

4. The composition of claim 1, wherein: the effective amounts of anti-infectious phytochemicals comprise two or more of the following:

Radix bupleuri extract;

Echinacea purpurea extract;

sage extract;

Uncaria tomentosa extract;

Black Elderberry extract;

American Elderberry; or

pomegranate extract; and

the effective amounts of anti-inflammatory phytochemicals comprise two or more of the following:

curcumin extract with piperine;

green tea extracts;

resveratrol;

Commiphora myrrha extract;

Uncaria tomentosa extract; and

Pinus maritima bark extract.

9

5. The composition of claim 4, wherein the effective amount of zinc comprises zinc in any of its pyrrhione, citrate, gluconate, picolinate, sulfate, acetate, or orotate forms.

6. The composition of claim 4, wherein the effective amount of zinc comprises zinc with a cofactor of methionine.

7. The composition of claim 1, wherein:
the effective amount of vitamin C is 10 mg-2000 mg; and
the effective amount of vitamin E is 5 mg-1000 mg.

8. The composition of claim 5, wherein the effective amount of zinc, which comprises zinc in any of its pyrrhione, citrate, gluconate, picolinate, acetate, sulfate, or orotate forms, is 5 mg-50 mg.

9. The composition of claim 5, wherein the effective amounts of anti-infectious phytochemicals comprise two or more of the following:

10 micrograms-1500 mg *Echinacea purpurea*;

10 micrograms-3200 mg sage extract;

10 micrograms-2000 mg *Uncaria tomentosa* extract;

10 micrograms-2000 mg Black Elderberry extract;

10 micrograms-1750 mg pomegranate extract; and

10 micrograms-1000 mg *Radix bupleuri* extract.

10. The composition of claim 1, comprising alpha-lipoic acid.

10

11. The composition of claim 9, wherein:

(i) the effective amounts of anti-inflammatory nutraceuticals comprise:

10 micrograms to 900 mg of omega-3 fatty acids comprising eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), or both;

(ii) the effective amounts of anti-inflammatory phytochemicals comprise two or more of the following:

10 micrograms to 1,200 mg of curcumin extract with piperine;

10 micrograms to 900 mg of green tea extract;

10 micrograms to 1,800 mg of *Boswellia serrata* extract containing 3-acetyl-11-keto-beta-boswellic acid (AKBA);

10 micrograms to 500 mg of resveratrol;

10 micrograms to 750 mg of extract of *Pinus maritima* bark extract;

10 micrograms to 800 mg of *Commiphora myrrha* extract; and

an effective amount of *Uncaria tomentosa* extract; and wherein the composition further comprises 10 micrograms to 1,200 mg of alpha-lipoic acid.

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